

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

Gal Barak

github.com/GalBrk

Nitzan Zacharia

github.com/NitzanZacharia

Inbal Moryles

github.com/iinbal

Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. We evaluate seven primer conditions against four controls jointly — a graph-blind shortcut solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and a degree-preserving rewiring — across four Qwen3 arms (1.7B and 4B, with and without thinking mode), six GraphQA tasks and seven edge densities on 40-node Erdős–Rényi graphs. A primer’s effect reverses across task, density and model size: the same `degree` primer is worth +26.5 points against the length control on `edge_count` and −6.0 on `node_degree`, where it states the requested number outright. What the primer changes is which procedure the model runs. Given a list of degrees, the share of responses that sum them and halve rises from 7% to 100%; where the primer states the answer, the median response collapses from 273 to 34 tokens and accuracy falls from 99 to 87. Accuracy then depends on where in the 40-line primer the queried node sits, by up to 21.6 points; and bundling three primers into one recovers a median 0.44 of what they are worth separately, losing gains and damage alike. Reading any of this requires two controls that are not usually applied: a length-matched content-free preamble, which is itself worth up to 15.7 points, and balanced accuracy on the boolean task, where raw accuracy rises with density while the model converges on answering “yes” to 98–99% of pairs.

1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask

a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

Four confounds complicate the interpretation of such an effect. First, a primer may help because the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task — or one whose task has a majority class large enough to score well by itself — leaves no room for any primer effect to appear, or credits one that is not there.

We ask what a primer does, and the headings of Section 5 are the questions in order. Which primer moves which task? What does the primer change in the model’s response — how much it writes, and which procedure it states? Where in the primer does it read? Do primers combine? And which of the answers hold across model arms and corpora? Section 5.1 comes first and is prerequisite rather than a result: at $n=40$ two of the six tasks have a majority class that scores well by itself, one is bounded by the generation budget, and a content-free preamble of matched length is already worth up to 15.7 points, so what counts as a primer effect has to be fixed before any of it can be read.

Two contributions support the answers. The first is the four controls of Figure 1 applied jointly — a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched filler, a minimal-length components, and a degree-preserving rewiring (Section 3.3). The second is that we read the generated text as well as the score, which is what lets us say what a primer did rather than only how large its effect was.

2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our *filler* condition measures this cost directly. The *rwse* condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

3 Method

3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (Figure 1; a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so they share sentence framing and number formatting and differ only in the facts stated. The exception is *filler*, which states no fact and therefore uses its own frame (“Node N is simply present within the graph G .”); it is a length control, not a content control. *rwse* states each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)).

3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact

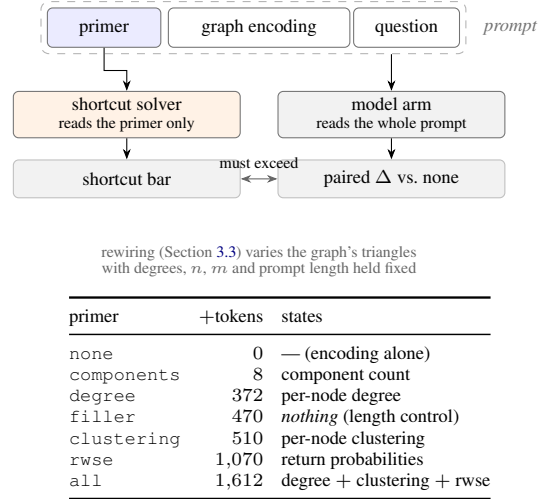


Figure 1: Experimental design. One prompt, two readers: the graph-blind shortcut solver sees only the primer and sets the bar a primer’s gain must exceed; the model arm sees all of it. Lengths are means at $n=40$ (Qwen3 tokenizer) against a mean *none* prompt of 2,165 tokens, so *filler* is a close length control for *clustering* but bounds only the length term for *rwse* and *all*.

rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Its accuracy on a (task, condition) cell, the *shortcut bar*, is a content-only lower bound: it never overstates what the primer text alone permits.

Bars are fit at the evaluated graph size: $n=40$ Erdős–Rényi graphs at the seven densities of the sweep, with disjoint 300-graph fit and test sets per density (Table 11); the screened arms, which answer the published 5–19-node split, use bars fit on that split, which differ substantially (Appendix Table 15). A (task, condition) pair offers a *substitute route* when its bar exceeds the *none* bar by more than 0.05; every gain is either at most 0.011 or at least 0.066, so any threshold between these values yields the same split. Only *node_degree* (under *rwse*, *degree*, *all*) and *edge_count* (under *clustering*, *degree*, *all*) offer a route. *node_count* and *cycle_check* have constant gold answers at $n=40$ (40, and “yes” at every density) and are excluded.

3.3 Rewiring: isolating structure from length

filler controls length only approximately, and comparisons across graph generators confound structure with class balance, so we use a within-instance design. A double-edge swap replaces edges $(a, b), (c, d)$ with $(a, d), (c, b)$ ([Fosdick](#)

et al., 2018), which preserves the degree sequence, n , m , the `node_degree` gold answer and the prompt length while changing the triangle count that the `clustering` primer reports. We generate low, base and high variants of each graph ($n=40$, mean degree 12, 200 graphs per level; mean clustering 0.01, 0.31, 0.73) and evaluate `node_degree` under `none`, `clustering` and `filler` with `qwen3-1.7b`, `qwen3-1.7b-think` and `qwen35-2b`. If the `clustering` benefit derives from the stated values, it should vary with triangle count. If instead the model reads mean clustering as a density prior (in an Erdős–Rényi graph $\bar{C} \approx p$), its degree estimates should shift toward $\bar{C}(n-1)$: 0.5, 12.0 and 28.7 across levels, against a true mean of 12.

3.4 Answer extraction and measurement validity

Extraction errors can masquerade as condition effects, so answers are extracted with task-specific patterns rather than a positional rule: on `cycle_check` a last-yes/no rule is not sufficient, because `qwen3-4b` often answers correctly and then recites that a cycle has “no repeated” nodes, which a positional rule scores against one condition more than another. A manual check of 50 rows per arm confirmed that `<think>` blocks never reach the extractor. Truncated pairs are dropped; scoring them as incorrect changes the verdict on only three `qwen3-1.7b` `cycle_check` cells (Appendix A).

3.5 Significance testing

Every comparison is paired on the same graph and question (Dror et al., 2018) and tested with the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected (Benjamini and Hochberg, 1995) within each (arm, task) family at $q=0.05$; a claim that a condition beats both controls also requires significance under a global correction. For the effect–baseline relation (Section 5.6), baseline and effect are estimated from disjoint random halves of each cell’s graphs (Chernozhukov et al., 2018), and the route $\times$ baseline interaction is bootstrapped over (arm, task, density) blocks, which share one `none` sample. For null cells we simulate a minimum detectable effect (MDE) at power 0.8, $\alpha=0.05$.

4 Experimental Setup

Tasks and arms. The six GraphQA tasks (Fatemi et al., 2023) are those named in Table 1. An *arm* is a model checkpoint with a fixed decoding configuration, so native thinking mode off and on gives two arms of one checkpoint.

Why $n=40$. On the published GraphQA split (Fatemi et al., 2023) — 5–19 nodes, five tasks — eight arms from two families score 0.97–1.00 under `none` on four of the five tasks (Appendix Table 13), leaving no headroom for a primer to move. We therefore evaluate on graphs large and dense enough to leave room in both directions.

Main sweep. Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025) carry it, each with thinking off (`qwen3-1.7b`, `qwen3-4b`) and on (`-think`): two sizes, both modes of the same weights, and the two sizes that retain headroom on five of six tasks at $n=40$. All four arms answer one shared prompt set of Erdős–Rényi graphs (Erdős and Rényi, 1959) with $n=40$ at densities $p \in \{.10, .20, .35, .50\}$, 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm, extended on `node_degree` and `edge_existence` to $p \in \{.65, .75, .85\}$ (100 graphs per cell, all four arms) so that the density range reaches from near-ceiling to near-floor. Decoding is greedy with a budget of 2,048 new tokens for plain arms and 8,192 for thinking arms; Section 5.6 reports a budget-matched comparison. Every headline result below is measured on this sweep.

Supporting corpora. Four further corpora each answer one question the main sweep cannot, and each is used only for that question. (i) A dedicated `node_degree` sweep with `qwen3-1.7b` and `qwen3-1.7b-think` — 400 graphs per density at all seven densities, with a `filler` arm, plus an independent replication on 400 newly seeded graphs per density at $p \leq .50$ — supplies the power the main sweep lacks for the one non-answer content effect (Section 5.6). (ii) A degree-preserving rewiring corpus (Section 3.3; `qwen3-1.7b`, `qwen3-1.7b-think` and Qwen3.5-2B (Qwen Team, 2026), `qwen35-2b`) tests whether that effect tracks the triangle count the primer reports. (iii) A fixed-mean-degree grid — mean degree ≈ 8 or ≈ 16 with $n \in \{20, 40, 80, 160\}$ on `node_degree`, answered by `qwen3-1.7b` and `qwen3-8b` — separates answer size from graph size and tests the same effect across four graph

sizes (Section 5.6); it carries no `filler` arm. (iv) A 100-graph probe of `none` and `degree` on all six tasks, answered by Qwen3-0.6B with and without thinking and by qwen3-2b, adds a third family and a smaller size to the robustness estimate in Appendix Table 20.

Screened arms. The eleven arms of the starting point and the probe support the two checks of Section 5.6; their shortcut bars are refit on their own corpus (Appendix Table 15).

Metrics. Exact match for every task; mean absolute error for the integer tasks is in the Appendix. On `connected_nodes` the choice of metric decides whether an effect can appear at all: set-F1, the metric that task is usually reported on, awards partial credit on a large answer set and sits at 0.93–1.00 in every cell, while exact match on the same responses spans 0.48–0.92. We report exact match throughout. On `edge_existence` we report balanced accuracy alongside accuracy, because the share of node pairs that are joined rises from 10% to 85% across our density range (Section 5.1).

5 Results

5.1 What these tasks and scores measure

A primer can only move accuracy where accuracy can move, and on two of the six tasks the score moves for reasons that have nothing to do with the primer. This section fixes what the numbers in the rest of the paper mean.

Accuracy on `edge_existence` tracks the class prior. The share of node pairs that are joined rises with density, from 10% at $p=.10$ to 85% at $p=.85$, so a constant answer earns a score that changes with density. Under `none`, qwen3-1.7b scores 89, 73, 66, 54, 68, 77, 86 across the seven levels — a decline and then an apparent recovery. Balanced accuracy over the same responses runs 93.9, 83.5, 75.4, 55.8, 52.9, 52.1, 53.3: the recovery is entirely the prior. Above $p=.50$ the model answers “yes” to 98–99% of pairs, and always saying yes scores 66–85% there. The same split holds on qwen3-4b, less severely (raw 96 to 96, balanced 97.8 to 89.4). We read no gain above $p=.50$ as evidence about graph reading (Appendix Figure 3).

Two tasks have no label distribution at all. At $n=40$ every one of the 400 graphs contains a

cycle and has 40 nodes, so `cycle_check` and `node_count` have a majority-class baseline of 1.000. `cycle_check` is uninformative here. `node_count` is worse: qwen3-1.7b scores 0.315, because 68.5% of its answers read the encoder’s “nodes 0, ..., 39” preamble as the count. Almost any interposed text interrupts that off-by-one, which is what the large primer gains on this task measure.

Truncation is an outcome, not a missing value. qwen3-1.7b reaches the output budget on 22.4% of `edge_count` generations; the thinking arms reach it on 81.0% and 62.3%. It decides scores elsewhere too: on `cycle_check`, whose answer is “yes” for all 400 graphs, qwen3-1.7b scores 0.997 on the generations that terminate and 0.923 counting the rest as wrong. Paired differences below drop a pair when either side truncated, the least favourable convention for the primer; rates are in Table 16.

Difficulty is set by the answer, not the graph. Holding mean degree fixed while n ranges over $\{20, 40, 80, 160\}$, growing n eightfold costs qwen3-1.7b 29 points while doubling $\bar{d}$ costs 33–42 at every n (Appendix Table 12). The sharper version is available within the main sweep, because one graph per (density, index) serves every task: `node_degree`, `connected_nodes` and `edge_existence` ask about the *same node of the same graph* and differ only in whether the answer is a count, a set or a boolean. Under `none` at $p=.50$, qwen3-1.7b scores 0.30, 0.48 and 0.54 on the three; at $p=.35$, 0.41, 0.59 and 0.66. Counting a neighbour set is harder than writing it out — and qwen3-4b reverses that ordering (0.99 against 0.79 at $p=.50$), so the required output operation is a difficulty driver in its own right, with graph, density and target all controlled, but which operation is harder depends on the model.

Added text is an active treatment on its own. `filler` is length-matched and states nothing about the graph, and it is not a null: it significantly reduces accuracy on `connected_nodes` and `edge_existence` in both plain arms. The cost is not a flat charge for tokens. It differs by task by a factor of two (Appendix Table 14a); within a task it is not monotone in density, running $-5, -20, -32, -6, -2, -1, -1$ from $p=.10$ to $.85$ on qwen3-1.7b/`edge_existence`, largest in the middle and gone where the task collapses to-

ward a constant answer; and it changes sign *within* every task but the constant-gold `node_count`, so there is no per-token rate to quote and no transferring an estimate from one task to another. Whether it tracks difficulty depends on the arm: correlated with accuracy under `none` it gives +0.55 for `qwen3-1.7b-think`, -0.09 for `qwen3-1.7b` and -0.44 for `qwen3-4b`. Every difference below is therefore reported against `filler` as well as against `none`.

On `edge_existence` these are shifts in response bias rather than in how the graph is read: the hit rate stays at 98–100% under every condition, so accuracy moves only through the false-alarm rate, which `filler` raises and `clustering` lowers. With hits at ceiling, sensitivity and criterion cannot be separated (Macmillan and Creelman, 2005; Hautus, 1995) (Appendix Tables 17 and 18).

5.2 Which primer moves which task?

Tables 1 and 1 give the main sweep against both baselines. Three things are visible in them that a single headline would hide.

The sign is not a property of the primer. `degree` is worth +26.5 points against the length control on `qwen3-4b/edge_count` and -6.0 on the same arm’s `node_degree`, where the primer states the requested number outright. `clustering` is worth +11.0 on `qwen3-4b/edge_existence` and nothing on that arm’s `node_degree`. Between arms, `degree` on `node_degree` runs +8.0 on `qwen3-1.7b` and -6.0 on `qwen3-4b`. Effects reverse across task, across density and across model size, so no primer is good or bad in itself (Appendix Figure 5).

Effects are close to one-directional. The paired design decomposes each difference into items the primer fixed and items it broke, and the two are rarely both large. At $p=.50$, `degree` on `qwen3-4b/edge_count` fixes 33 items and breaks none; `rwse` on that arm’s `edge_existence` breaks 23 and fixes none; `degree` on its `node_degree` breaks 13 and fixes 1. `clustering` is the exception, fixing 8 and breaking 4 on `edge_existence`, so its small net is a different kind of effect from the others rather than a weaker version of the same one, and should not be read as one.

The largest content effect we measure costs eight tokens. `components` on `connected_nodes` gains +9.2 points over `none` and +21.8 over `filler` for `qwen3-4b` ($n=400$, $p \approx 10^{-4}$, both under the global correction; Table 1). Every condition beats `filler` there, because `filler` itself costs 12.5 points; what distinguishes `components` is that it also beats `none`. Since it adds 8 tokens where `filler` adds 470 (Figure 1), it is the only effect we measure that is substantially free of length cost.

Carrying the right feature is neither necessary nor sufficient. Each primer reports one graph feature and each task needs one, so some (primer, task) pairs are matched and the rest are not. The three largest gains we measure are matched pairs — `degree` on `edge_count` (+26.0 on `qwen3-4b`), `components` on `connected_nodes` (+21.7) and `clustering` on `edge_existence` (+19.5 on `qwen3-1.7b`). But mismatched content is still worth +8.1 points on `qwen3-1.7b`, more than two of the four matched pairs, and the fourth matched pair reverses with model size. Relevance organises the largest effects without predicting the rest (Appendix Table 10).

5.3 What does the primer change in the response?

Accuracy alone cannot say what a primer did. The generated text can, and it shows the primer changing which procedure the model runs and how much work it does, in directions that differ by task.

Before the primer, `edge_count` is budget-bound. Under `none` the plain arms almost never answer exactly — `qwen3-4b` is exact on 8 of 400 with a median relative error of 10.9% and never truncates, `qwen3-1.7b` exact on 5 with 28.5% over the 70% of its generations that terminate — while the thinking arms are exact on 86% and 98% of the generations that terminate and truncate on 82% and 80% of all of them. The task is not beyond the plain arms’ competence; without an explicit serial count they substitute an estimate, and with one they run out of budget.

On `edge_count` the primer replaces the method. Under `none` at $p=.50$, 7% of `qwen3-1.7b`’s responses state the handshake shortcut — sum the degrees and divide by two — and the rest enumerate edges. Under

Arm	Task	none	Δ vs none						Δ vs filler				
			filler	comp.	clust.	rwse	degree	all	comp.	clust.	rwse	degree	all
qwen3-1.7b	node_count [†]	31.5	+39.8	+21.2	+67.3	+2.0	+6.8	+47.0	-18.5	+27.5	-37.8	-33.0	+7.3
	cycle_check [†]	99.7	+0.3	-0.6	+0.3	-7.5	-12.6	-1.1	-0.9	+0.0	-7.6	-11.7	-1.1
	edge_existence	70.5	-15.7	-8.2	+3.8	+5.3	+4.0	+11.5	+7.5	+19.5	+21.1	+19.8	+27.2
	node_degree	60.5	-0.5	+1.8	+3.0	+4.0	+7.5	+10.3	+2.3	+3.5	+4.8	+8.0	+10.8
	connected_nodes	72.5	-6.2	+0.8	+1.3	+1.0	+3.5	+2.0	+7.0	+7.5	+7.3	+9.8	+8.3
	edge_count	1.8	-0.4	-1.4	-0.4	-0.8	+4.0	+2.4	-0.4	-0.4	-0.4	+4.5	+1.3
qwen3-4b	node_count [†]	90.2	+9.3	+9.0	+8.8	+9.3	+8.0	+9.8	-0.2	-0.5	+0.0	-1.2	+0.5
	cycle_check [†]	99.7	-0.3	-1.3	+0.3	+0.3	-1.5	-0.5	-1.3	+0.3	+0.3	-1.3	-0.5
	edge_existence	90.5	-7.0	-2.8	+4.0	-10.0	-4.3	-9.8	+4.2	+11.0	-3.0	+2.5	-2.8
	node_degree	99.2	-0.5	+0.2	-0.5	-2.3	-6.5	-8.0	+0.7	+0.0	-1.8	-6.0	-7.5
	connected_nodes	86.0	-12.5	+9.3	+1.0	-0.0	-3.0	-2.7	+21.8	+13.5	+12.5	+9.5	+9.8
	edge_count	2.0	+1.2	-1.3	-1.3	-1.5	+27.7	+3.8	-2.5	-2.5	-2.8	+26.5	+2.5
qwen3-1.7b-think	node_count [†]	98.6	+1.4	+0.9	+1.4	+1.4	+0.8	+1.4	-0.3	+0.0	+0.0	-0.8	+0.0
	cycle_check [†]	100.0	+0.0	+0.0	+0.0	+0.0	-0.7	+0.0	+0.0	+0.0	+0.0	-0.7	+0.0
	edge_existence	99.2	-0.8	-0.8	-1.0	-1.3	-2.3	-2.0	+0.0	-0.3	-0.5	-1.5	-1.2
	node_degree	76.8	-3.5	-1.0	+1.0	+2.5	+11.7	+11.6	+2.3	+4.5	+5.8	+14.6	+15.1
	connected_nodes	79.7	+2.0	+2.8	+1.5	-0.3	+7.4	+6.1	+0.5	-0.5	-2.5	+4.6	+4.1
	edge_count	86.3	+3.4	+7.4	-8.7	-4.0	-61.5	-8.3	+5.4	-4.2	+13.0	-18.8	-22.2
qwen3-4b-think	node_count [†]	99.5	+0.5	+0.0	+0.5	+0.5	+0.5	+0.5	-0.5	+0.0	+0.0	+0.0	+0.0
	cycle_check [†]	100.0	+0.0	+0.0	+0.0	+0.0	-0.8	+0.0	+0.0	+0.0	+0.0	-0.8	+0.0
	edge_existence	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_degree	99.7	+0.0	-0.3	-0.3	-2.1	+0.0	-0.8	-0.5	-0.3	-2.1	-0.3	-0.8
	connected_nodes	98.7	+0.8	+0.5	+0.0	+0.0	+0.0	-0.5	+0.0	+0.0	-0.5	-0.5	-0.5
	edge_count	97.6	-2.0	+1.7	+1.7	+0.0	-23.5	-41.5	+2.3	+0.0	-6.3	-19.3	-36.4

Table 1: Main sweep, $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The `none` column is exact-match accuracy (%); the rest are paired differences in percentage points, against `none` on the left and against the length-matched `filler` on the right, with truncated generations excluded pairwise. Reading a primer against `none` alone charges it for the text it adds; reading it against `filler` alone hides what that text costs, which the `filler` column gives. [†]: gold answer is constant at $n=40$, so a shift need not reflect graph reading (Section 5.1). **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family and baseline, $q = 0.05$.

`degree`, which lists all 40 degrees, 100% state it. Truncation falls from 29% to 20% and the median terminating response from 3,737 to 818 new tokens. The same switch appears at every density, and at $p=.10$ runs 67% to 99%. This is a wholesale change of algorithm, not a nudge: the primer supplies an arithmetic route around an enumeration the model cannot finish inside its budget. `filler` induces the switch in 31% of responses and `all` in 40%, so stating the degrees is what carries it.

Where the primer hurts, reasoning collapses.

On `qwen3-4b/node_degree` at $p=.50$, where the `degree` primer states the requested number verbatim, the median response falls from 273 new tokens under `none` to 34, and accuracy falls from 99 to 87 — the primer breaks 13 items and fixes 1. Neither cell truncates at all, so this is a choice and not a budget limit. `filler` leaves the median at 266 tokens and costs 2 points, so it is the stated answer and not the added text that shortens the

response. The model abandons a reliable 273-token count for an unreliable 34-token lookup.

The direction is task-specific. On `qwen3-1.7b/edge_existence` the same primers make the model do *more*: 206 tokens under `none` against 294 under `degree` and 352 under `all`. On `qwen3-4b/connected_nodes`, `filler` alone cuts the median from 233 to 94 while costing 12.5 points — a content-free preamble that makes the model answer faster and worse. “Primers make responses longer” and “primers make responses shorter” are both false as general statements (Appendix Table 9).

At high density the smaller model stops reasoning altogether. On `edge_existence` beyond $p=.50$, `qwen3-1.7b`’s responses shorten sharply and converge on the majority answer: it replies “yes” to 98–99% of pairs and its balanced accuracy sits at chance (Appendix Figure 3). Neither prompt length nor task difficulty triggers this. A ladder that raises n at fixed mean degree makes the

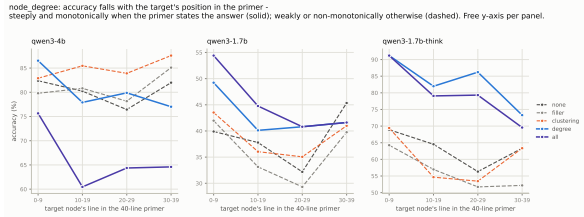


Figure 2: `node_degree` accuracy by the queried node’s line number within the 40-line primer, pooled over all seven densities ($n=700$ per condition). Primers that state the answer show the steepest gradient; the controls, which contain no per-node value to retrieve, are flatter but not flat.

prompt longer and the task harder without producing it, and the collapse appears on the task whose answer space is a single token while the enumeration tasks on the same graphs keep generating (Appendix Figure 4). What the two-valued answer format permits is a cheap guess that is right most of the time, and raw accuracy rewards it.

5.4 Where in the primer does the model read?

If a primer is being retrieved from rather than reasoned over, accuracy should depend on *where* in the 40 lines the queried node sits. It does, and the dependence is large. We fit correctness on the target’s line number and report the slope as points from the first line to the last, with a percentile bootstrap interval over the ≈ 700 items in each cell; quartile means are in Figure 2. On `qwen3-1.7b-think`, `all` loses 24.3 points $[-33.5, -15.2]$ from the top of the primer to the bottom and `degree` loses 18.7 $[-27.7, -9.7]$; on `qwen3-4b`, `degree` loses 12.1 $[-22.3, -3.0]$. For comparison, the whole effect of the `degree` primer on that arm is -6.0 points. Where in the primer the answer sits matters more than whether the primer is there.

Node identity does not explain it. Position and node id are the same variable here, because the primer lists nodes in id order, so the finding would collapse if later-numbered nodes were simply harder nodes. They are not: over the 19,600 `node_degree` items, a node’s id correlates $+0.019$ with its degree and $+0.020$ with its clustering coefficient. Erdős–Rényi ids are assigned before edges are drawn, so the 40th line of the primer describes a node no harder than the first.

But retrieval does not explain all of it either.

Two cells rule that out. `rwse` on `qwen3-1.7b` has the steepest slope we measure, -19.3 $[-31.5, -7.3]$, and it never prints the queried degree; and `filler` on `qwen3-1.7b-think` loses 16.4 points $[-29.0, -4.0]$ while containing no per-node content whatsoever. A preamble that says nothing about any node still makes the model worse at nodes described late in it. Averaged over the four arms the ordering is in the expected direction — -12.3 points for the primers that state the answer, -5.1 for per-node primers that do not, -2.8 for the controls — and under `none` no arm has an interval excluding zero. So position costs most where there is a value to find, but the cost is not only the cost of finding it.

What we cannot say from this design is which cost it is. The slope is consistent with retrieval failing further into a list, with a general degradation over prompt position that any long preamble induces, or with both; separating them needs a primer whose line order is shuffled away from node-id order, which we did not run. Two further limits: the four quartiles are too coarse to distinguish monotone decay from a lost-in-the-middle shape, and the slope does not grow monotonically with density $(-9, -12, -24, -24, -12, -33, -18)$ across the seven levels for `degree` on `qwen3-1.7b-think`), so we do not claim it scales with how much there is to search.

5.5 Do primers combine?

`all` renders `degree`, `clustering` and `rwse` in one primer, so the parts and the bundle are both measured on the same graphs. Bundling is lossy: across the 48 cells where the parts do anything, the bundle delivers a median 0.44 of their sum (Table 2). How lossy depends on how cells are weighted — the estimate runs from 0.29 to 0.71 — so what the data support is that the bundle falls well short of its parts, not a coefficient. The shortfall is the same whether the parts help or hurt: `qwen3-1.7b/edge_existence` keeps 0.90 of a $+12.8$ gain and `qwen3-4b/edge_existence` keeps 0.93 of a -10.8 loss. That symmetry is what dilution of a useful sentence among irrelevant ones would predict and not what interference would. `edge_count` is the exception: there the bundle keeps 0.15 of a $+24.5$ gain on `qwen3-4b` and moves against its parts on two thinking arms. It is also the task whose gain depends on a single stated

Arm	Task	$\sum$ parts	all	ratio
<i>other tasks</i>				
1.7b	connected_nodes	+5.7	+2.0	0.35
1.7b	edge_existence	+12.8	+11.5	0.90
1.7b	node_degree	+14.5	+10.2	0.71
1.7b-t	connected_nodes	+8.8	+6.2	0.71
1.7b-t	edge_existence	-4.5	-2.0	0.44
1.7b-t	node_degree	+16.5	+12.5	0.76
4b	edge_existence	-10.8	-10.0	0.93
4b	node_degree	-9.2	-8.0	0.86
<i>edge_count</i>				
1.7b-t	edge_count	-8.0	+9.0	-1.12
4b	edge_count	+24.5	+3.7	0.15
4b-t	edge_count	+35.2	-6.7	-0.19
<i>median over cells</i>				0.44

Table 2: all renders degree, clustering and rwse in one primer, so the parts and the bundle are both measured. The first two columns are paired differences against none in percentage points, averaged over $p \in \{0.10, 0.20, 0.35, 0.50\}$; rows whose parts sum to under 4 points are omitted, since a ratio of two null effects is not informative. The final row is the median over the 48 individual (arm, task, density) cells rather than over the averaged rows above, because averaging numerator and denominator over densities first gives a ratio of means that the largest cells dominate; weighted the other ways the estimate runs from 0.29 to 0.71, so it bounds a tendency and is not a coefficient. Either way the bundle delivers well under the sum of its parts, and the shortfall is the same whether the parts help or hurt, which is what dilution rather than interference would predict. edge_count is the extreme case: there the bundle keeps almost nothing and in two arms moves against its parts.

procedure, which the bundle buries.

5.6 What holds across arms and corpora?

Answer-stating primers track the baseline. Where a primer offers a substitute route to the answer, it gains on cells with low accuracy under none and loses on cells with high accuracy, while primers that offer no such route show no such relation. Estimated on cross-fitted halves of each cell’s graphs, so that baseline and effect do not share sampling noise, the interaction is -11.0 points per unit of baseline (95% CI $[-17.1, -5.7]$; Appendix Figure 6 and Table 20). It holds within node_degree alone, which truncates on 0–1.6% of generations, so it is not an artifact of censoring, and it has the same sign on the eleven screened arms. It is a contrast between floor and ceiling rather than a graded dependence: restricted to baselines between 0.10 and 0.90 it disappears.

One non-answer content effect replicates.

clustering on node_degree is the only effect of a primer that neither states the answer nor merely pays for its length that we could replicate on a dedicated corpus. Pooled over $p=.35$ and .50 it exceeds none by +5.9 points (95% CI $[2.1, 9.8]$) and filler by a similar margin, with filler and none indistinguishable there; an independent-seed replication gives +5.5. On the fixed-degree grid of Appendix Table 12 it gains +6.2 points ($n=3,191$; $p<10^{-4}$), larger where the answer is larger. Neither mechanism we can test explains it: under degree-preserving rewiring, mean clustering rises from 0.01 to 0.73 while the benefit stays flat, and the signed error shifts by less than 0.3 where a density-prior account predicts up to 17 (Appendix Figure 7 and Table 23).

Thinking mode buys accuracy partly with budget. Thinking brings most cells to ceiling, and where it does not, truncation censors the task. Rescored at the plain arms’ 2,048-token budget, which is exact under greedy decoding, the 1.7B thinking arm still wins on edge_existence (97.8 against 70.5%) but loses on cycle_check (38.0 against 90.7%; Appendix Table 24). Many of the main-sweep nulls are uninformative: none is already at floor or ceiling in 42% of qwen3-1.7b’s null cells and in all 33 of qwen3-4b-think’s. Where a gain could be detected, the median detectable size is 5.5 points for the 1.7B arms and 5.4 for 4B.

6 Conclusion

A primer’s effect on GraphQA accuracy is not a property of the primer. It reverses across task, across density and across model size, it is close to one-directional within a cell — fixing many items and breaking none, or the reverse — and a content-free preamble of the same length is itself worth up to 15.7 points, so any effect measured against none alone confounds content with length.

The generated text says what the primer did. A primer that lists the degrees replaces edge enumeration with an arithmetic shortcut in every response, cutting length and truncation with it; a primer that states the requested number collapses the response from 273 tokens to 34 and costs accuracy, breaking 13 items to fix 1. At high density the smaller model stops reasoning on the two-valued task altogether and guesses the majority class, which raw accuracy rewards and balanced accuracy does not.

What the effects depend on is partly legible and partly not. Position inside the primer matters, by up to 21.6 points, though a length-matched control also slopes on one arm. Carrying the feature a task needs organises the largest effects but predicts neither the rest nor the sign. Bundling three primers recovers a median 0.44 of their separate worth, equally whether they help or hurt.

All claims are behavioural. Evaluations of prompt-level graph augmentations should report a length-matched control, a graph-blind solver, a majority-class baseline for every task, and enough of the generated text to say which procedure changed.

Limitations

The main sweep covers two model sizes from one family; the screened arms extend the range but on a smaller corpus with fewer conditions and were not pre-registered. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the `clustering` primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The length control matches `clustering` to within 8% in tokens but is much shorter than `rwse` and `all`. Every prompt uses one encoding (`incident`), one template and one primer position, with greedy decoding. Several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample.

Three limits bear on the behavioural results specifically. The solution strategy a response states is identified by regular expressions over free text, hand-validated on a stratified sample, and a model may use a procedure it does not narrate. Position within the primer is confounded with node identity, because the primer’s line order is node-id order; the `none` and `filler` controls bound this, but `filler` itself slopes on one arm, so we can say the gradient is steepest under answer-stating primers and not that it is purely retrieval. The response-length collapse is attributed to the answer format by eliminating prompt length and task difficulty, not by manipulating the format directly.

Four further conditions would settle the main open questions: *shuffled-value* (a stricter placebo than `filler`), *corrupted* (to separate copying from computation), *single-fact* (to separate dilution from interference in the bundle) and a *reordered*

primer (to break the position/identity confound above).

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

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A Supplementary material

A.1 Example prompt

A clustering-condition node_degree prompt at $p=0.10$ (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.

Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

A.2 Per-task results

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	31.5	+39.8	+21.2	+67.3	+2.0	+6.8	+47.0
4b	90.2	+9.3	+9.0	+8.8	+9.3	+8.0	+9.8
1.7b-t	98.6	+1.4	+0.9	+1.4	+1.4	+0.8	+1.4
4b-t	99.5	+0.5	+0.0	+0.5	+0.5	+0.5	+0.5

Table 3: node_count, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at $n=40$ on this task, so a shift need not reflect graph reading. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	99.7	+0.3	−0.6	+0.3	−7.5	−12.6	−1.1
4b	99.7	−0.3	−1.3	+0.3	+0.3	−1.5	−0.5
1.7b-t	100.0	+0.0	+0.0	+0.0	+0.0	−0.7	+0.0
4b-t	100.0	+0.0	+0.0	+0.0	+0.0	−0.8	+0.0

Table 4: cycle_check, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at $n=40$ on this task, so a shift need not reflect graph reading. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	70.5	-15.7	-8.2	+3.8	+5.3	+4.0	+11.5
4b	90.5	-7.0	-2.8	+4.0	-10.0	-4.3	-9.8
1.7b-t	99.2	-0.8	-0.8	-1.0	-1.3	-2.3	-2.0
4b-t	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0

Table 5: `edge_existence`, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The `none` column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	60.5	-0.5	+1.8	+3.0	+4.0	+7.5	+10.3
4b	99.2	-0.5	+0.2	-0.5	-2.3	-6.5	-8.0
1.7b-t	76.8	-3.5	-1.0	+1.0	+2.5	+11.7	+11.6
4b-t	99.7	+0.0	-0.3	-0.3	-2.1	+0.0	-0.8

Table 6: `node_degree`, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The `none` column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	72.5	-6.2	+0.8	+1.3	+1.0	+3.5	+2.0
4b	86.0	-12.5	+9.3	+1.0	+0.0	-3.0	-2.7
1.7b-t	79.7	+2.0	+2.8	+1.5	-0.3	+7.4	+6.1
4b-t	98.7	+0.8	+0.5	+0.0	+0.0	+0.0	-0.5

Table 7: `connected_nodes`, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The `none` column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	1.8	-0.4	-1.4	-0.4	-0.8	+4.0	+2.4
4b	2.0	+1.2	-1.3	-1.3	-1.5	+27.7	+3.8
1.7b-t	86.3	+3.4	+7.4	-8.7	-4.0	-61.5	-8.3
4b-t	97.6	-2.0	+1.7	+1.7	+0.0	-23.5	-41.5

Table 8: `edge_count`, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The `none` column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

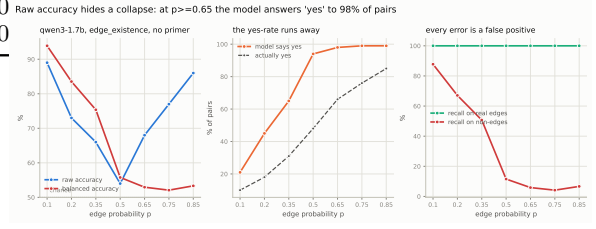


Figure 3: `edge_existence` under `none`, qwen3-1.7b. Accuracy falls to 54% at $p=.50$ and then rises again to 86% at $p=.85$, which reads as a recovery. Balanced accuracy over the same responses does not recover: it reaches chance at $p=.65$ and stays there, because the model answers “yes” to 98–99% of pairs while 66–85% of them really are edges.

Arm / task		none	filler	degree	all
1.7b edge_count	median tokens	—	—	818	—
	% sum-of-deg.	7%	31%	100%	40%
	% truncated	29%	41%	20%	44%
4b node_degree	median tokens	273	266	34	134
	% sum-of-deg.	0%	0%	0%	0%
	% truncated	0%	0%	0%	0%
1.7b edge_existence	median tokens	206	171	294	352
	% sum-of-deg.	0%	0%	1%	0%
	% truncated	0%	0%	0%	0%
4b connected_nodes	median tokens	233	94	258	107
	% sum-of-deg.	0%	0%	0%	0%
	% truncated	0%	0%	0%	0%

Table 9: Response behaviour at $p=0.50$. Median new tokens is computed over generations that terminated, since a capped response is truncated at the budget rather than long, and is reported as — where more than 20% of generations truncated, because the surviving median is then a sample of the short responses; read the truncation row for those cells. “sum-of-deg.” is the share of responses that state the handshake shortcut (sum the degrees, divide by two) rather than enumerating edges. The primer moves length in both directions depending on the task, and on `edge_count` it replaces the method outright.

Primer / task	1.7b	4b	1.7b-t	4b-t
<i>matched: the primer carries what the task needs</i>				
clustering / edge_existence	+19.5	+11.0	-0.2	+0.0
components / connected_nodes	+7.0	+21.7	+0.2	-0.8
degree / edge_count	+3.5	+26.0	-5.5	-3.5
degree / node_degree	+8.0	-6.0	+15.2	+0.5
<i>mismatched, mean over all pairs</i>				
—	+8.1	+2.0	+1.5	-0.5

Table 10: Each primer reports one graph feature and each task needs one, so a pair is *matched* when the primer carries what the task asks for. Entries are paired differences against *filler* in percentage points, averaged over $p \in \{0.10, 0.20, 0.35, 0.50\}$; constant-gold tasks are excluded. The three largest gains we measure are all matched pairs, but matching is not sufficient and not necessary: mismatched content is still worth +8.1 points on qwen3-1.7b, more than two of the matched pairs, and the fourth matched pair reverses with model size. That pair is degree on node_degree, the one case where the primer states the requested number outright: it gains 8 points on the 1.7B arm and loses 6 on the 4B arm (Section 5.4).

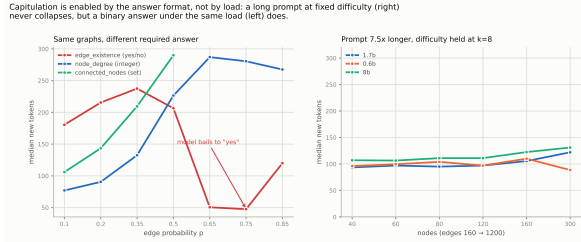


Figure 4: The response-length collapse on *edge_existence* and two controls on it. (a) median new tokens by density under *none* for three tasks answered about the *same* graphs: only the task with a two-valued answer collapses, while the tasks that must enumerate keep generating, so the trigger is not the graph. (b) median new tokens against n at fixed mean degree, where the prompt grows and the task gets harder without producing a collapse. What the collapse tracks is the answer format.

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 11: Shortcut bars at $n=40$ (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

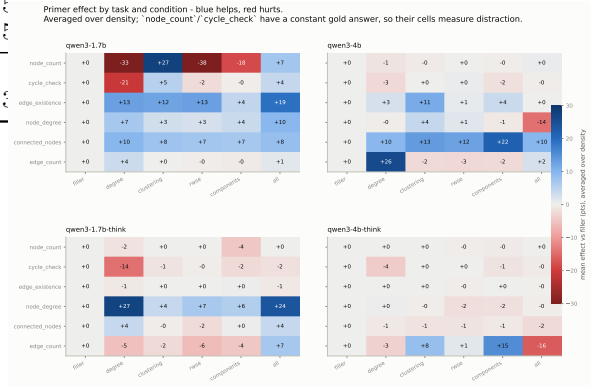


Figure 5: Every cell of Table 1 as a heat map: mean paired effect against *filler* by task and primer, one panel per arm, on a shared symmetric scale so that zero is the same colour throughout. The point of the figure is that the sign is not a property of a primer: rows and columns both change colour.

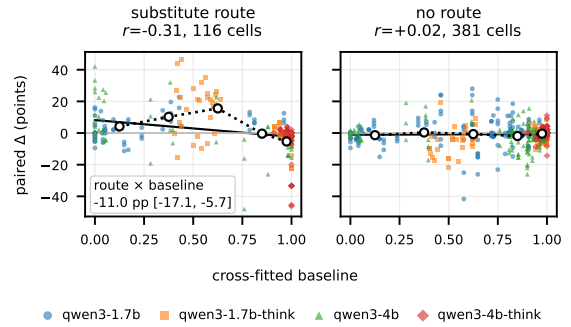


Figure 6: Cross-fitted paired effect against cross-fitted baseline accuracy (Section 5.6) for the 116 cells in which a primer offers a substitute route to the answer (left) and the 381 that do not (right), each shown once per fold direction. Baseline and effect come from disjoint halves of the paired graphs, so shared sampling noise cannot produce the slope.

n	$\bar{d}$	qwen3-1.7b			qwen3-8b	
		none	clust.	Δ	none	Δ
20	8	0.912	0.865	-4.8*	0.998	+0.2
40	8	0.728	0.770	+4.2	0.995	+0.0
80	8	0.645	0.750	+10.5*	0.965	-0.8
160	8	0.624	0.693	+6.9*	0.897	-2.8
20	16	0.575	0.740	+16.5*	0.985	-0.2
40	16	0.398	0.476	+7.8*	0.975	+1.0
80	16	0.225	0.323	+9.8*	0.895	-1.8
160	16	0.224	0.211	-1.3	0.802	-4.0
pooled		0.542	0.604	+6.2	$n=3191, p<10^{-4}$	

Table 12: Fixed-mean-degree sweep on `node_degree`: mean degree $\bar{d}$ is held while n varies, so density falls as the prompt lengthens. Growing n eightfold at $\bar{d}\approx 8$ costs 29 points and saturates after $n=80$; doubling $\bar{d}$ costs 33–42 points at every n . Δ is clustering against none, exact match. *: Benjamini–Hochberg at $q=0.05$ across the eight cells. This corpus has no filler arm, so Δ is a net effect. qwen3-8b is a ceiling control.

A.3 Response behaviour

A.4 Route substitution

A.5 Robustness analyses

Arm	node_count	node_degree	edge_count	connected_nodes	cycle_check	all	cap
gemma4-12b	1.00	1.00	0.93	1.00	1.00	0.987	0/150
gemma4-12b-think	1.00	1.00	1.00	1.00	1.00	1.000	43/150
gemma4-e4b	1.00	1.00	0.96	1.00	0.97	0.986	5/150
gemma4-e4b-think	0.97	1.00	0.93	1.00	0.97	0.973	0/150
qwen3-8b	1.00	1.00	0.43	1.00	0.97	0.880	0/150
qwen3-8b-think	1.00	1.00	1.00	1.00	1.00	1.000	7/150
qwen3-14b	0.97	1.00	0.40	1.00	0.97	0.867	0/150
qwen3-14b-think	1.00	1.00	1.00	0.97	1.00	0.993	2/150

Table 13: The published GraphQA split (5–19 nodes, 30 graphs per cell) under none, exact match. Every arm scores 0.97–1.00 on four of the five tasks; only `edge_count` separates them, and only for the two plain Qwen arms (0.40–0.43 against 0.93–1.00 for the rest). `cap` counts generations that hit the token budget, excluded from the accuracies. This corpus cannot measure a primer effect, which is why the main sweep moves to $n=40$.

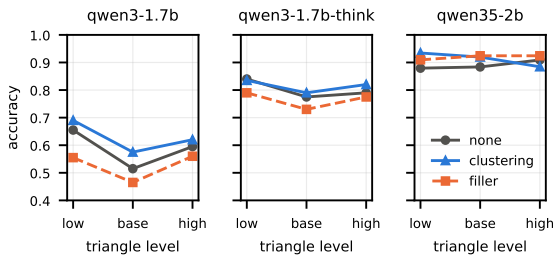


Figure 7: Rewiring: `node_degree` accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

(a) length cost (filler–none) by task			
Task	cells	mean	range
edge_existence	11	-9.1	$[-32.0, -1.0]$
connected_nodes	7	-6.3	$[-14.0, +3.1]$
node_degree	11	-4.3	$[-17.2, +4.0]$
node_count	1	+31.0	$[+31.0, +31.0]$
(b) content gain (cond.–filler) by arm			
Cond.	1.7b / 1.7b-think / 4b		
components	+5.7 / +6.8 / +5.5		
clustering	+10.9 / +2.6 / +11.6		
rwse	+11.3 / +5.6 / +5.2		

Table 14: Mid-range cells. (a) The cost of added text is task-specific, not a per-token rate, and changes sign within tasks, so it cannot be transferred across them. `node_count` rests on a single cell and is a constant-gold task; it is shown for completeness only. (b) Only `components` holds its content gain across arms; the ranking of the other two reverses between 1.7b and 4b, so no single ordering of primers by length survives disaggregation.

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946

Table 15: Shortcut bars on 500 graphs from the published 5–19-node generator, used for the screened arms. At $n=40$ (Table 11) the `edge_existence/none` bar rises to 0.735 through class imbalance, and `degree`’s gap over none on that task falls from +0.296 to +0.011.

A.6 Per-cell results

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, an exact 95% interval conditional on the discordant pairs (Clopper–Pearson, inverting the same exact McNemar test used for the per-cell p -values), the permutation p -value, and Δ under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 3.4). n is the number of pairs in which neither arm truncated. “n/a” marks a cell in which no pair disagreed: the conditional estimate is then exactly zero with no width, which is a statement about this sample and not a bound on the effect, so the discordant count is reported instead.

Task	1.7B	1.7B-th.	4B	4B-th.
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 16: Truncation rate (share of main-sweep generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density. **Bold**: above 50%.

Arm	Cond.	hit rate	false alarm
qwen3-1.7b	none	100.0%	40.3%
	filler	100.0%	61.8%
	clustering	100.0%	35.2%
qwen3-4b	none	99.1%	12.6%
	filler	100.0%	22.5%
	clustering	100.0%	7.5%

Table 17: edge_existence, pooled over the four main-sweep densities: hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

Cond.	qwen3-1.7b				qwen3-4b			
	$\Delta d'$	Δc	$\Delta d'$	Δc	$\Delta d'$	Δc	$\Delta d'$	Δc
components	-0.36 [-0.56, -0.21]	-0.18	+0.07 [-0.24, +0.33]	-0.10	+0.21 [+0.03, +0.41]	+0.11	+0.43 [+0.12, +0.75]	+0.08
clustering	+0.07 [-0.19, +0.43]	+0.28	-0.38 [-0.68, -0.15]	-0.32	+0.15 [-0.05, +0.41]	+0.19	-0.04 [-0.33, +0.21]	-0.15
degree	-0.93 [-1.16, -0.71]	-0.46	-0.25 [-0.54, +0.00]	-0.25	+0.55 [+0.35, +0.77]	+0.27	-0.36 [-0.71, -0.11]	-0.31
filler								
all								

Table 18: edge_existence, main sweep: change in sensitivity d' and criterion c against none, averaged over the four densities (log-linear correction; 95% bootstrap CI for $\Delta d'$). With the hit rate at ceiling, d' and c are not separately identifiable.

Accuracy under none	substitute route		no route	
	cells	Δ	cells	Δ
0.00–0.25	24	+6.3	24	-0.7
0.25–0.50	6	+8.5	6	+2.0
0.50–0.75	6	+15.5	24	+0.4
0.75–0.95	11	+4.2	33	-3.9
0.95–1.00	34	-9.0	186	-0.4
$r(\Delta, \text{baseline})$	-0.417		-0.050	

Table 19: Mean primer effect (percentage points), binned by none accuracy on the same (arm, task, density, condition) cell of the main sweep, using the $n=40$ route split (Table 11). Mean baseline is 0.61 for route cells and 0.85 for no-route cells, so the groups are not baseline-matched; the cross-fitted interaction test in the text addresses this.

Estimate	r_{route}	k	r_{none}	k	interaction [95% CI]
naive, pp	-0.35	117	-0.01	381	-12.5 [-19.5, -6.3]
cross-fit (one direction), pp	-0.34	116	+0.03	381	-12.1 [-17.9, -7.0]
cross-fit (both), pp	-0.31	232	+0.02	762	-11.0 [-17.1, -5.7]
cross-fit (both), logit	-0.40	232	+0.04	762	-1.35 [-1.93, -0.77]
baseline .10-.90, pp	-0.01	78	-0.07	187	+3.0 [-29.1, +41.1]
baseline .10-.90, logit	+0.02	78	-0.05	187	+0.22 [-1.36, +1.94]
screened, pp	-0.67	105	-0.19	93	-24.7 [-109, -3.8]

Table 20: Route $\times$ baseline relation, main sweep plus high-density extension (screened: published split and probe). “Both” counts each cell once per fold direction; k for the window rows counts points. Interaction CIs resample (arm, task, density) blocks, or (arm, task) for screened arms.

Comparison	Δ	n	density	p
clust. vs. none (main)	+3.0	400	.10-.50	0.26
clust. vs. none	+1.9	2800	.10-.85	0.022
filler vs. none	-2.4	2800	.10-.85	0.0013
clust. vs. filler	+4.3	2800	.10-.85	2.4×10^{-8}
clust. vs. none	+7.5	400	.35	0.0089
clust. vs. none (new seed)	+4.2	1600	.10-.50	0.00055

Table 21: qwen3-1.7b, node_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 22.

p	clust.-none	filler-none	clust.-filler
.10	+0.3	+3.0*	-2.8*
.20	+3.3	+0.5	+2.8
.35	+7.5*	+1.0	+6.5*
.50	+4.3	-2.0	+6.3*
.65	-2.0	-9.5*	+7.5*
.75	+0.5	-6.3*	+6.8*
.85	-0.5	-3.5*	+3.0*

Table 22: qwen3-1.7b, node_degree, dedicated sweep, 400 graphs per density. *: McNemar $p < 0.05$ (uncorrected).

Arm	low	base	high
predicted shift	-11.5	0.0	+16.7
qwen3-1.7b	+0.03	+0.28	-0.02
qwen3-1.7b-think	+0.01	+0.04	+0.01
qwen35-2b	-0.29	-0.26	-0.16

Table 23: Density-prior test on the rewiring corpus: mean paired shift in signed node_degree error, clustering minus none (errors clipped at ± 20 , 200 graphs per level). “Predicted” is the shift if answers moved fully to $\bar{C} (n - 1)$.

Task	1.7B, acc. under none (%)			4B, acc. under none (%)		
	plain	think	think@2048	plain	think	think@2048
connected_nodes	72.5	78.7	67.2	86.0	95.5	89.0
cycle_check	90.7	97.0	38.0	99.0	99.3	62.0
edge_count	1.3	15.8	0.0	2.0	20.0	0.0
edge_existence	70.5	99.0	97.8	90.5	99.8	99.8
node_count	31.5	91.0	86.0	90.2	99.5	94.8
node_degree	60.5	76.2	67.7	99.3	97.0	91.2

Table 24: Budget-matched comparison. Plain arms use 2,048 tokens; thinking arms 8,192 (“think”) or rescored at 2,048 (“think@2048”). Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Table 1.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
1.7B	node_count	0.69	0.47	0.01	0.67	0.62	0.29	0.22
	edge_count	133.4	98.1	105.7	79.0	47.7	181.2	94.2
	node_degree	0.94	0.91	0.78	0.91	0.67	1.27	0.91
4B	node_count	0.10	0.01	0.01	0.01	0.02	0.01	0.00
	edge_count	39.2	40.1	43.4	56.0	40.4	29.3	28.1
	node_degree	0.02	0.01	0.02	0.05	0.14	0.02	0.49

Table 25: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	400	+0.8	[−3.1, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[−3.3, +5.7]	0.6458	+1.2
	rwse	400	+1.0	[−3.5, +5.5]	0.7304	+1.0
	degree	400	+3.5	[−0.8, +7.5]	0.1239	+3.5
	filler	400	−6.2	[−10.0, −1.9]	0.0040	−6.2
	all	399	+2.0	[−2.9, +6.8]	0.4644	+2.0
cycle_check	components	343	−0.6	[−0.6, +0.4]	0.5033	+2.0
	clustering	362	+0.3	[−0.3, +0.3]	1.0000	+8.5
	rwse	358	−7.5	[−8.1, −5.2]	0.0001	−0.2
	degree	301	−12.6	[−12.6, −10.3]	0.0001	−17.5
	filler	341	+0.3	[−0.3, +0.3]	1.0000	+3.0
	all	362	−1.1	[−1.6, +0.5]	0.2210	+7.0
edge_count	components	211	−1.4	[−2.3, +1.0]	0.3783	−1.0
	clustering	238	−0.4	[−2.4, +1.9]	1.0000	−0.5
	rwse	239	−0.8	[−2.3, +1.4]	0.6837	−0.8
	degree	253	+4.0	[+0.6, +5.8]	0.0179	+3.0
	filler	224	−0.4	[−2.0, +1.6]	1.0000	−0.5
	all	211	+2.4	[−1.1, +4.6]	0.2269	+0.8
edge_existence	components	400	−8.2	[−10.9, −4.7]	0.0001	−8.2
	clustering	400	+3.8	[−0.9, +8.1]	0.1199	+3.8
	rwse	399	+5.3	[+0.4, +9.8]	0.0321	+5.0
	degree	400	+4.0	[−0.7, +8.4]	0.0957	+4.0
	filler	400	−15.8	[−18.3, −12.0]	0.0001	−15.8
	all	400	+11.5	[+6.5, +15.8]	0.0001	+11.5
node_count	components	400	+21.2	[+15.9, +25.6]	0.0001	+21.2
	clustering	400	+67.2	[+65.4, +67.2]	0.0001	+67.2
	rwse	400	+2.0	[−4.5, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+2.9, +9.9]	0.0008	+6.8
	filler	400	+39.8	[+37.2, +40.6]	0.0001	+39.8
	all	400	+47.0	[+43.8, +48.7]	0.0001	+47.0
node_degree	components	400	+1.8	[−2.4, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[−2.0, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[−0.9, +8.6]	0.1104	+4.0
	degree	400	+7.5	[+2.8, +11.6]	0.0018	+7.5
	filler	400	−0.5	[−4.6, +3.7]	0.9028	−0.5
	all	400	+10.2	[+4.9, +15.0]	0.0003	+10.2

Table 26: qwen3-1.7b: all cells.

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	392	+2.8	[−0.7, +5.9]	0.1268	+2.5
	clustering	394	+1.5	[−2.7, +5.6]	0.5384	+2.0
	rwse	393	−0.3	[−4.3, +3.8]	1.0000	+0.2
	degree	390	+7.4	[+3.5, +10.6]	0.0001	+6.2
	filler	394	+2.0	[−1.7, +5.5]	0.3275	+2.5
	all	392	+6.1	[+2.2, +9.4]	0.0027	+6.2
cycle_check	components	346	+0.0	n/a (0 disc.)	1.0000	−7.5
	clustering	376	+0.0	n/a (0 disc.)	1.0000	−0.5
	rwse	386	+0.0	n/a (0 disc.)	1.0000	+2.2
	degree	295	−0.7	[−0.7, +0.5]	0.5036	−21.8
	filler	388	+0.0	n/a (0 disc.)	1.0000	+3.0
	all	373	+0.0	n/a (0 disc.)	1.0000	−0.8
edge_count	components	27	+7.4	[−5.1, +7.4]	0.5037	−0.8
	clustering	23	−8.7	[−17.2, +10.6]	0.6251	−1.2
	rwse	25	−4.0	[−4.0, +3.8]	1.0000	−3.2
	degree	13	−61.5	[−61.5, −16.1]	0.0102	−3.2
	filler	29	+3.4	[−8.4, +10.2]	1.0000	−1.5
	all	24	−8.3	[−22.8, +13.9]	0.6794	+5.0
edge_existence	components	398	−0.8	[−1.9, +0.9]	0.5090	−0.8
	clustering	398	−1.0	[−1.9, +0.6]	0.2911	−1.0
	rwse	398	−1.3	[−2.4, +0.6]	0.2274	−1.2
	degree	397	−2.3	[−3.1, −0.3]	0.0249	−2.5
	filler	399	−0.8	[−1.9, +0.9]	0.5051	−0.5
	all	399	−2.0	[−2.9, −0.1]	0.0374	−1.8
node_count	components	342	+0.9	[−0.6, +1.4]	0.3758	+1.5
	clustering	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
	rwse	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
	degree	360	+0.8	[−0.8, +1.8]	0.4512	+5.5
	filler	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
	all	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
node_degree	components	395	−1.0	[−4.4, +2.6]	0.6543	−1.0
	clustering	396	+1.0	[−3.0, +4.9]	0.7010	+1.5
	rwse	394	+2.5	[−1.9, +6.8]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +15.0]	0.0001	+8.8
	filler	396	−3.5	[−7.3, +0.6]	0.0954	−3.0
	all	396	+11.6	[+7.3, +15.0]	0.0001	+12.2

Table 27: qwen3-1.7b-think: all cells.

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	400	+9.2	[+6.1, +11.3]	0.0001	+9.2
	clustering	400	+1.0	[−3.1, +5.0]	0.6893	+1.0
	rwse	400	+0.0	[−4.2, +4.2]	1.0000	+0.0
	degree	400	−3.0	[−7.1, +1.4]	0.1882	−3.0
	filler	400	−12.5	[−16.2, −7.9]	0.0001	−12.5
	all	400	−2.8	[−7.0, +1.7]	0.2511	−2.8
cycle_check	components	391	−1.3	[−1.8, +0.3]	0.1318	−2.0
	clustering	395	+0.3	[−0.2, +0.3]	1.0000	+0.5
	rwse	392	+0.3	[−0.2, +0.3]	1.0000	−0.2
	degree	391	−1.5	[−1.5, −0.1]	0.0312	−2.2
	filler	392	−0.3	[−0.3, +0.2]	1.0000	−0.5
	all	395	−0.5	[−1.0, +0.6]	0.6269	−0.2
edge_count	components	397	−1.3	[−2.1, +0.5]	0.1756	−1.2
	clustering	399	−1.3	[−2.4, +0.6]	0.2298	−1.2
	rwse	398	−1.5	[−2.0, +0.1]	0.0668	−1.5
	degree	393	+27.7	[+24.5, +29.4]	0.0001	+27.3
	filler	400	+1.2	[−0.9, +2.9]	0.3012	+1.2
	all	397	+3.8	[+1.1, +5.6]	0.0049	+3.8
edge_existence	components	400	−2.8	[−5.8, +0.7]	0.1254	−2.8
	clustering	400	+4.0	[+0.7, +6.7]	0.0168	+4.0
	rwse	400	−10.0	[−12.9, −6.2]	0.0001	−10.0
	degree	398	−4.3	[−7.3, −0.7]	0.0200	−4.8
	filler	400	−7.0	[−10.0, −3.3]	0.0002	−7.0
	all	399	−9.8	[−12.9, −5.8]	0.0001	−10.0
node_count	components	400	+9.0	[+6.9, +9.5]	0.0001	+9.0
	clustering	400	+8.8	[+6.2, +9.9]	0.0001	+8.8
	rwse	400	+9.2	[+6.9, +10.1]	0.0001	+9.2
	degree	400	+8.0	[+5.1, +9.7]	0.0001	+8.0
	filler	400	+9.2	[+7.5, +9.2]	0.0001	+9.2
	all	400	+9.8	[+8.0, +9.8]	0.0001	+9.8
node_degree	components	400	+0.2	[−0.2, +0.2]	1.0000	+0.2
	clustering	400	−0.5	[−1.7, +1.0]	0.7246	−0.5
	rwse	400	−2.2	[−3.1, −0.3]	0.0240	−2.2
	degree	400	−6.5	[−7.7, −4.0]	0.0001	−6.5
	filler	400	−0.5	[−1.7, +1.0]	0.7232	−0.5
	all	400	−8.0	[−8.9, −5.6]	0.0001	−8.0

Table 28: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	$[-0.6, +1.0]$	0.6249	+0.5
	clustering	373	+0.0	$[-1.2, +1.2]$	1.0000	-0.8
	rwse	379	+0.0	$[-1.4, +1.4]$	1.0000	+0.5
	degree	376	+0.0	$[-1.5, +1.5]$	1.0000	+0.5
	filler	379	+0.8	$[-0.6, +1.3]$	0.3787	+1.2
	all	375	-0.5	$[-1.8, +1.1]$	0.7306	-1.2
cycle_check	components	381	+0.0	n/a (0 disc.)	1.0000	-3.2
	clustering	394	+0.0	n/a (0 disc.)	1.0000	+0.0
	rwse	395	+0.0	n/a (0 disc.)	1.0000	+0.2
	degree	375	-0.8	$[-0.8, +0.3]$	0.2514	-5.5
	filler	396	+0.0	n/a (0 disc.)	1.0000	+0.5
	all	392	+0.0	n/a (0 disc.)	1.0000	-0.5
edge_count	components	60	+1.7	$[-1.6, +1.7]$	1.0000	+24.8
	clustering	59	+1.7	$[-4.1, +5.0]$	1.0000	+24.8
	rwse	49	+0.0	$[-4.0, +4.0]$	1.0000	+10.2
	degree	34	-23.5	$[-23.5, -6.1]$	0.0073	+8.5
	filler	50	-2.0	$[-2.0, +1.9]$	1.0000	+12.2
	all	41	-41.5	$[-41.5, -25.3]$	0.0002	+1.8
edge_existence	components	396	+0.0	n/a (0 disc.)	1.0000	-0.5
	clustering	396	+0.0	n/a (0 disc.)	1.0000	-0.8
	rwse	393	+0.0	n/a (0 disc.)	1.0000	-1.2
	degree	398	+0.0	n/a (0 disc.)	1.0000	+0.0
	filler	399	+0.0	n/a (0 disc.)	1.0000	+0.0
	all	394	+0.0	n/a (0 disc.)	1.0000	-1.0
node_count	components	400	+0.0	$[-0.9, +0.9]$	1.0000	+0.0
	clustering	399	+0.5	$[-0.3, +0.5]$	0.4936	+0.2
	rwse	399	+0.5	$[-0.3, +0.5]$	0.5005	+0.2
	degree	399	+0.5	$[-0.3, +0.5]$	0.5005	+0.2
	filler	400	+0.5	$[-0.3, +0.5]$	0.4870	+0.5
	all	400	+0.5	$[-0.3, +0.5]$	0.4870	+0.5
node_degree	components	380	-0.3	$[-0.3, +0.3]$	1.0000	+0.2
	clustering	386	-0.3	$[-0.8, +0.6]$	1.0000	+1.8
	rwse	381	-2.1	$[-2.1, -0.5]$	0.0072	-1.2
	degree	388	+0.0	$[-0.5, +0.5]$	1.0000	+2.5
	filler	386	+0.0	n/a (0 disc.)	1.0000	+2.0
	all	383	-0.8	$[-0.8, +0.3]$	0.2466	+0.5

Table 29: qwen3-4b-think: all cells.